

Varun Maram

Postdoctoral Research Fellow

Cybersecurity Group

SandboxAQ

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Research Interests

My interests lie in *quantum-resistant cryptography*, with an emphasis on provable post-quantum security of real-world cryptosystems. I also have a broader interest in *multi-party computation* and *distributed systems*. Recently, I've been interested in constructing advanced quantum-secure applications – such as anonymous credentials and verifiable oblivious PRFs – from zkSNARKs.

Education

- 2019 - **Swiss Federal Institute of Technology, Zurich (ETH Zurich), Zurich**
- 2023 Dr.Sc. in Computer Science, member of the Applied Cryptography Group
Supervisor: Kenneth G. Paterson

- 2017 - **Swiss Federal Institute of Technology, Zurich (ETH Zurich), Zurich**
- 2019 M.Sc. in Computer Science “with distinction”, GPA – **5.75/6.0**

- 2013 - **Indian Institute of Technology, Roorkee (IIT Roorkee), Roorkee**
- 2017 B.Tech. in Computer Science and Engg., minors in Mathematics, CGPA – **9.578/10**
(Departmental Rank 2, out of ≈ 75 students)

Work Experience

- 03/2024 - **SandboxAQ, London**
- Present *Research Scientist – Postdoc.* Member of the Cybersecurity Group.
Manager(s): Nina Bindel, Andreas Hülsing
 - Currently analysing post-quantum versions of real-world protocols, and constructing advanced quantum-secure applications from zkSNARKs.

- 06/2022 - **Visa Research, Palo Alto**
- 09/2022 *Research Intern.* Worked on constructing quantum secure public-key encryption schemes. Mentor: Navid Alamati

- 06/2021 - **Visa Research, Palo Alto (worked remotely from Zurich)**
- 09/2021 *Research Intern.* Worked on analysing quantum (in)security of widely-used block cipher modes of operation. Mentors: Daniel Masny and Sikhar Patranabis

- 05/2016 - **Adobe BigData Experience Lab, Bangalore**
- 07/2016 *Research Intern.* Conducted research on augmented reality technology in the context of digital marketing. Mentor: Gaurush Hiranandani

Standardization Efforts

- 2022 D. J. Bernstein, T. Chou, C. Cid, J. Gilcher, T. Lange, [Varun Maram](#), I. von Maurich, R. Misoczki, R. Niederhagen, E. Persichetti, C. Peters, N. Sendrier, J. Szefer, C. J. Tjhai, M. Tomlinson, and W. Wang. **“Classic McEliece”**, Round 4 Submission in NIST’s *Post-Quantum Cryptography (PQC) Standardization Project*, currently being considered for ISO standardization.

Publications

- 2025 N. Alamati, [Varun Maram](#). **“Quantum CCA-Secure PKE, Revisited”**, *Journal of Cryptology – JoC, Volume 38*.
- 2025 B. Benčina, B. Dowling, [Varun Maram](#), and K. Xagawa. **“Post-quantum Cryptographic Analysis of SSH”**, *46th IEEE Symposium on Security and Privacy 2025*. [\[This work was also presented at the Real World Crypto \(RWC\) 2025 symposium.\]](#)
- 2024 N. Alamati, [Varun Maram](#). **“Quantum CCA-Secure PKE, Revisited”**, *27th IACR Intl. Conference on Practice and Theory of Public-Key Cryptography – PKC 2024*. [\[Best Paper Award\]](#), [\[Invited to Journal of Cryptology\]](#)
- 2023 M. Jauch, [Varun Maram](#). **“Quantum Cryptanalysis of OTR and OPP: Attacks on Confidentiality, and Key-Recovery”**, *Selected Areas in Cryptography – SAC 2023*.
- 2023 N. Alamati, [Varun Maram](#), D. Masny. **“Non-Observable Quantum Random Oracle Model”**, *14th Intl. Conference on Post-Quantum Cryptography – PQCrypto 2023*.
- 2023 [Varun Maram](#), K. Xagawa. **“Post-Quantum Anonymity of Kyber”**, *26th IACR Intl. Conference on Practice and Theory of Public-Key Cryptography – PKC 2023*. [\[Best Paper Award\]](#)
- 2022 [Varun Maram](#), D. Masny, S. Patranabis, and S. Raghuraman. **“On the Quantum Security of OCB”**, *IACR Transactions on Symmetric Cryptology, ToSC 2022 (2)*.
- 2022 P. Grubbs, [Varun Maram](#), and K. G. Paterson. **“Anonymous, Robust Post-Quantum Public Key Encryption”**, *41st Annual Intl. Conference on the Theory and Applications of Cryptographic Techniques – EUROCRYPT 2022*. [\[This work was also presented at NIST’s 3rd PQC Standardization Conference.\]](#)
- 2021 K. Cong, D. Cozzo, [Varun Maram](#), and N. P. Smart. **“Gladius: LWR Based Efficient Hybrid Public Key Encryption with Distributed Decryption”**, *27th Annual Intl. Conference on the Theory and Application of Cryptology and Information Security – ASIACRYPT 2021*.
- 2020 [Varun Maram](#). **“On the Security of NTS-KEM in the Quantum Random Oracle Model”**, *8th Intl. Workshop on Code-Based Cryptography – CBCrypto 2020*.
- 2020 C. Liu-Zhang, [Varun Maram](#), and U. Maurer. **“On Broadcast in Generalized Network and Adversarial Models”**, *24th Conference on Principles of Distributed Systems – OPODIS 2020*.

- 2019 C. Liu-Zhang, *Varun Maram*, and U. Maurer. “**Brief Announcement: Towards Byzantine Broadcast in Generalized Communication and Adversarial Models**”, 33rd International Symposium on Distributed Computing – *DISC 2019*.
- 2017 G. Hiranandani, K. Ayush, A. R. Sinha, *Sai Varun Reddy Maram*, C. Varsha, and P. Maneriker. “[**Poster**] **Enhanced Personalized Targeting Using Augmented Reality**” 16th IEEE Intl. Symposium on Mixed and Augmented Reality – *ISMAR 2017*.

--- Selected Talks

- 2025 “**Post-quantum Cryptographic Analysis of SSH**”
- Real World Crypto Symposium – *RWC 2025*
 - Secure Shell Maintenance (SSHM) Working Group Session – *IETF 123*
- 2024 “**Don’t Roll Your Own FO: Challenges in Proving Post-Quantum CCA Security of Real-World KEMs**”
- Workshop on Key Exchange and Channel Protocols – *SKECH 2024*
- 2024 “**Quantum CCA-Secure PKE, Revisited**”
- Information Security Group (ISG) Research Seminar, Royal Holloway (RHUL)
 - 27th IACR Intl. Conference on Practice and Theory of PKC – *PKC 2024*
- 2023 “**Post-Quantum Anonymity of Kyber**”
- CWI Amsterdam Crypto Group Seminar
 - 26th IACR Intl. Conference on Practice and Theory of PKC – *PKC 2023*
- 2023 “**Non-Observable Quantum Random Oracle Model**”
- 14th Intl. Conference on Post-Quantum Cryptography – *PQCrypto 2023*.
- 2023 “**On the Quantum Security of OCB**”
- Fast Software Encryption Conference – *FSE 2023*
- 2022/23 “**Anonymous, Robust Post-Quantum Public Key Encryption**”
- IBM Research Zurich, Security Department Seminar
 - ENS Lyon/CWI Amsterdam/RHUL Joint Online Cryptography Seminar
 - QSIT/Quantum Centre, ETH Zurich, Lunch Seminar
 - NIST’s Third PQC Standardization Conference
 - 41st Annual Conference on Cryptographic Techniques – *EUROCRYPT 2022*.
- 2020 “**On the Security of NTS-KEM in the Quantum Random Oracle Model**”
- 8th Intl. Workshop on Code-Based Cryptography – *CBCrypto 2020*.
- 2020 “**On Broadcast in Generalized Network and Adversarial Models**”
- 24th Conference on Principles of Distributed Systems – *OPODIS 2020*.
- 2019 “**Brief Announcement: Towards Byzantine Broadcast in Generalized Communication and Adversarial Models**”
- 33rd International Symposium on Distributed Computing – *DISC 2019*.

Selected Honors and Awards

- 2024 **Best Paper Award** given by the program committee of PKC 2024 conference for our paper “*Quantum CCA-Secure PKE, Revisited*”; given to 2 papers out of 178 submissions.
- 2023 **Best Paper Award** given by the program committee of PKC 2023 conference for our paper “*Post-Quantum Anonymity of Kyber*”; given to 2 papers out of 183 submissions.
- 2017 **Master Scholarship** awarded by the Department of Computer Science, ETH Zurich to admitted students based on their excellent academic achievements so far; was offered to ≈ 10 students that year.
- 2014 **Aditya Birla Group Scholarship** awarded to engineering, management and law students all over India, based on their excellence on the academic and leadership front; given to ≈ 40 graduates in the country.
- 2012 **Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship** awarded by Dept. of Science and Technology, Govt. of India, to highly motivated students interested in pursuing a research career; given to ≈ 250 high school students in the country.
- 2011 **Bronze Medal**, 16th International Astronomy Olympiad, Kazakhstan. One of the 3 high school students selected to represent India in the competition.
- 2011 **Infosys Award** for International Olympiad Medallists, by HBCSE in association with Tata Institute of Fundamental Research (TIFR) Endowment Fund
- 2011 **Gold Medal**, 13th National Science Olympiad, Science Olympiad Foundation, Delhi. National Rank 1st out of $\approx 20,000$ participants.

Teaching

I was a teaching assistant for the following courses at the Department of Computer Science, ETH Zurich.

- 2022-2023 **Zero-Knowledge Proofs** (*Master's level*)
- 2023 **Seminar on Current Topics in Cryptography** (*Master's level*)
- 2022 **Information Security** (*Bachelor's level*)
- 2020-2021 **Applied Cryptography** (*Master's level*)
- 2020-2021 **Algorithmic Game Theory** (*Master's level*)

Advising

I supervised the following theses at the Department of Computer Science, ETH Zurich.

- 2023 Melanie Jauch. “**Quantumania: Three Quantum Attacks on AES-OTR's Confidentiality and a Quantum Key-Recovery Attack on OPP**” (*Bachelor's thesis*)
[This work was also published at the SAC 2023 conference.]

- 2022 Marc Himmelberger. **“Concrete IND-CCA Security of NIST PQC KEMs in the ROM”** (*Bachelor’s thesis*)
- 2021 Oliver Tran. **“Exploring RSA Assumptions”** (*Bachelor’s thesis*)

Academic Service

I was on the program committee for the following conference:

- *PKC 2025*

I was also a sub-reviewer for the following conferences:

- *EUROCRYPT 2026, PKC 2026*
- *CRYPTO 2025, ASIACRYPT 2025, TCC 2025*
- *CRYPTO 2024, ASIACRYPT 2024*
- *EUROCRYPT 2023*
- *CRYPTO 2021, ASIACRYPT 2021*
- *CRYPTO 2020, CT-RSA 2020*

Patents

- 2016 G. Hiranandani, Sai Varun Reddy Maram, K. Ayush, C. Varsha, and S. Jain. **“Method, Medium, and System for Product Recommendations Based on Augmented Reality Viewpoints”**, *US 10475103 B2*.
- 2016 G. Hiranandani, C. Varsha, Sai Varun Reddy Maram, K. Ayush, and A. R. Sinha. **“Identifying Augmented Reality Visuals Influencing User Behavior in Virtual-Commerce Environments”**, *US 10950060 B2*.
- 2016 G. Hiranandani, K. Ayush, C. Varsha, and Sai Varun Reddy Maram. **“Creating Targeted Content Based on Detected Characteristics of an Augmented Reality Scene”**, *US 10922716 B2*.

Technical Skills

Programming	C++, C, Python, Java, bash, C#
OS	Linux (Ubuntu), macOS, Windows
Software	Git, PostgreSQL, Docker, SageMath, liboqs, OpenSSH, EasyCrypt, LaTeX